

ANTIBIOTIC RESISTANCE PATTERN OF *SALMONELLA* FROM CLINICAL ISOLATES

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ABSTRACT:

Fifty three clinical isolates of *Salmonella* *S. typhi*, *S. para typhi A*, *S. para typhi B*) were collected from different pathological laboratories of Karachi. Resistance pattern of these isolates were determined against ampicillin, streptomycin, tetracycline, chloramphenicol and gentamicin. Seventy five percent of organisms were resistant to one or more antibiotics at 100 mg/ml concentration. The highest percentage of resistant isolates was shown against streptomycin (64%). Gentamicin was found to be most effective drug. The most common resistance pattern observed was Amp. sm. Tc.Cm. It was found out that the genetic location of these resistance markers is on resistant plasmid. This fact was proved by performing the experiment of curing and conjugation between *Salmonella* and *E.coli*. species.

INTRODUCTION

The genus *Salmonella* belongs to the large, medically important eubacterial family, "Enterobacteriaceae", whose natural habitat is the intestinal tract of humans and animals. *Salmonella* are ubiquitous pathogens of both warm-blooded and cold-blooded vertebrates (Shu-lin Liu and Sanderson, 1992). In contrast to the incidence of typhoid fever and many other infectious diseases in developed countries, the incidence of gastroenteritis due to *Salmonella* spp. has been steadily increasing for the past 30 years (Cohen and Tauxe, 1986; Rubin and Weinstein, 1977), where the hygienic conditions are much better than under developed countries.

Drug resistance is acquired in two ways. It may arise de novo by mutation or by gene amplification. It may be transferred to the infecting organism from bacteria in the form of extrachromosomal pieces of DNA that contain the information for the resistance mechanism. Most of the antibiotic resistance that occurs clinically is due to the transfer of segments of DNA or plasmid, containing drug resistance gene, from one bacterium to other. These are called R-plasmids and these can even be exchanged between cells of different species (Hardy, 1987). It seems that bacteria will evolve resistance to most if not all antibiotics with which they are challenged (Foster, 1983). The R-plasmids are very dangerous from epidemiologic point of view as the diseases caused by organisms bearing these R plasmids (by conjugation) will be difficult to control (Ansari and Khatoon, 1996).

Observation of the resistance of *Salmonella johannesburg* to six drugs: Amp.Sm.Tc.Cm.Km and Su were made over the seven years from 1973-1979. These resistances were found to be mediated by autotransferring plasmids belonging to the incompatibility group F1me (Chau et al., 1982). CDC conducted prospective studies of reported Salmonellosis. Frequency of resistance had increased particularly to tetracycline and chloramphenicol (MacDonald et al., 1986). Chloramphenicol has been the drug of choice

for decades. Other effective drugs include ampicillin and trimethoprim-sulfamethazole. If the frequency of resistance to these three antimicrobials rises, the choice of agents that can be used to treat patients becomes more limited (Tacket et al., 1985).

In this research work the study of several clinical isolates has been carried out for their resistance to commonly used antibiotics and it has been found that whether the resistance is plasmid-mediated or not.

MATERIALS AND METHODS

The cultures of *Salmonella* were collected from different pathological laboratories (Sindh Lab., Tariq Road, Dr. Essa's Laboratory, Dr. Ehsanullah's Laboratory, Liaquat National Hospital) of Karachi. In these pathological laboratories these cultures were isolated from stool and blood specimens of patients. *Salmonella* were selectively grown on Bismuth Sulphite agar (Difco). Cultures were purified on MacConkey agar plates. *Salmonella* were confirmed by observing Gram staining and reactions on Triple sugar iron agar (Difco). Cultures were maintained on MacConkey agar plate by sub culturing every month and kept in refrigerator.

10 mg/ml stock solutions of ampicillin, streptomycin, tetracycline, chloramphenicol and gentamicin were sterilized by 0.45 mm millipore filter and kept in freezer until used. Drug resistance was seen at 10, 25, 50, 100, 250 and 500 mg/ml concentrations by performing replica plating technique. Spots of different *Salmonella* cultures were made with the help of sterile tooth picks to make the master plate on MacConkey agar plate. This plate was incubated at 37°C. Next day the prints were made on different plates of MacConkey agar containing desired antibiotic concentration, with the help of replica wooden block. Plates were incubated overnight at 37°C. Appearance of colony showed the resistance and absence showed that the organisms are sensitive to that concentration of the antibiotic. In this way the resistance pattern was determined.

Curing was done by elevated temperature (Asheshov et al., 1966). The experiment was performed with *S. typhi* and *S. para typhi* A. Overnight broth cultures of the strains were diluted 1/4 in fresh tryptone soya broth and incubated at 37°C for one and half hours with aeration (in shaking water bath). Forty ml of this sample was added to each of the two tubes containing 5 mL tryptone soya broth, prewarmed to 37°C and to 43-44°C, respectively. The tubes were then incubated at appropriate temperature for five and half hours without aeration. The cultures were suitably diluted so that 0.1 mL contained 100-200 viable units and were spreaded on to tryptone soya agar plates and incubated overnight at 37°C. The resulting individual colonies were then picked for making a master plate, from where replicated on to MacConkey agar plate containing appropriate antibiotics to determine the curing effect on the characteristic resistance pattern of the isolates.

The experiment of conjugation was performed between *Salmonella* and *Escherichia coli* (Chau et al., 1982). The experiment was conducted with *Salmonella typhi* and *S. para typhi A* to *E. coli* FPL5014 and *E. coli* wild type respectively. The single colony of *Salmonella* and *E. coli* were inoculated to tryptone soya broth and incubated for overnight at 37°C. From this overnight culture 1.5 ml was transferred into a flask containing 8ml broth and kept in shaking water bath for 2 hours. When cloudy growth appeared, conjugation mixture was prepared. Donor control contained 1mL *Salmonella* culture and 1mL of tryptone soya broth. Recipient control contained 1mL each of *E. coli* and tryptone soya broth. Conjugation mixture contained 0.2mL of *Salmonella* culture and 1.8mL of *E. coli* culture. The cultures were mixed and incubated overnight (Iyer and Iyer, 1969). Next morning the tubes were centrifuged at 3000–4000 rpm for 20 minutes. The supernatant was discarded and the sediment was streaked on MacConkey agar plates added with antibiotics, so that only transconjugants would grow. We used two markers: lac positive and antibiotic resistance. The stability of the transfer of the resistance marker in these transconjugants was checked by replica plating on to antibiotic containing plates for 2-3 times.

RESULTS AND DISCUSSION

In the present research study resistance pattern of the 53 clinical isolates of *Salmonella* spp. was observed to five commonly used antibiotics i.e. ampicillin, streptomycin, tetracycline, chloramphenicol and gentamicin. Genetic location of resistance was determined by doing the experiment of curing. It was also found out that these R-plasmids are transferable.

Out of 53 cultures which were isolated from different patients with clinical symptoms, 41 (77%) were *S. typhi*, 9 (17%) were *S. para typhi A* and 3 (6%) were *S. para typhi B*. *Salmonella para typhi A* was found to be the most resistant (56%) followed by *S. typhi* (51%) and *S. para typhi B* (25%) at 100 mg/ml concentration. The most common drug resistance pattern was observed as Amp.Sm.Tc.Cm. (34%) among *Salmonella* isolates. Fifty percent of the isolates were resistant at 100 mg/ml concentration or more to different antibiotics. The highest number of resistance was found against streptomycin, to which *S. typhi* was 68%, *S. par typhi A* was 100%, and *S. para typhi B* was 67% resistant, at 100 mg/ml or more concentration. Out of 53 isolates 75% of organisms were resistant to one or more antibiotics at 100 mg/ml concentration (Table I). The cumulative resistance shown by *Salmonella* spp. against streptomycin was 39%, ampicillin 25%, tetracycline 22%, chloramphenicol 20% and gentamicin 0% (at 100 mg/ml concentration) (Fig. I). Therefore the most effective drug was found out to be gentamicin, as no resistance was seen against it, even at 10 mg/ml concentration, reason being the limited use of this antibiotic.

In order to confirm the locations (chromosomal or extra chromosomal) the strain of *S. typhi* was subjected to curing. The strain showed the loss of ampicillin, streptomycin and chloramphenicol resistance markers (Table II) and when the conjugation was performed with this strain to *E. coli* FPL5014 the recipient become resistant to ampicillin, streptomycin and

chloramphenicol at > 300 mg/ml concentration (Table III). In another experiment of conjugation with *S. para typhi* A to *E. coli* wild type the recipient become resistant to streptomycin, tetracycline and chloramphenicol at > 300 mg/ml concentration (Table IV). However we were unable to cure resistance markers in *S. para typhi* A, because some plasmids are not curable even by ethidium bromide or acridine orange (Amir and Khatoon, 1976; Khatoon, 1987).

Curing confirmed that the antibiotic resistance of the clinical isolates was due to the presence of resistant plasmid. The experiment of conjugation showed that these R-plasmids are transferable. The conjugative transfer of multiple antibiotic resistance markers was also observed. Chloramphenicol had been the drug of choice for salmonellosis (Cohen, 1986) but now resistance is developing against this antibiotic also. Ampicillin is another effective drug. If the frequency of resistance to these antimicrobials rises, the choice of agents to treat patients becomes more limited and the likelihood increases that the therapy may not be successful at all. The significance of antimicrobial resistance is most obvious in its impact on treatment of human infection.

The overuse and misuse of antimicrobials have lead to the death of sensitive strains; leaving the resistant strains to survive, multiply and infect new hosts. We have a tradition of prescribing antibiotics to any body who looks sick. The more widely you use newer antibiotics, the greater the chances that bacteria will develop resistance. Even when doctors dispense antibiotics properly, there is no guarantee they will be used that way. Many stop taking their medication on their own. The spread of antibiotic resistance is encouraged by the inappropriate use of antimicrobials, their being taken at wrong dosages and for an insufficient length of time. This is particular concern in developing countries where culture and sensitivity testing may not be carried out due to lack of resources. Mis-application of antibiotics and poor hygiene facilitate the spread of resistance. The selection of antibiotics is now becoming more limited as bacteria develop resistance to the available drugs and this poses an obvious to human health.

Table I
Resistance shown by *Salmonella* spp. to one or more antibiotics
(100 mg/ml)

Species	Sensitive to all antibiotics	Resistance of one antibiotics	Resistance to two antibiotics	Resistance to three antibiotics	Resistance to four antibiotics	% of Resistance
<i>S. typhi</i>	12	9	2	2	16	71%
<i>S. para typhi A</i>	0	4	1	2	02	100%
<i>S. para typhi B</i>	1	1	1	0	00	67%
Total	13	14	4	4	18	75%
Percentage	25%	26%	8%	8%	34%	

Percentage of resistance was calculated by number of isolates resistant to one or more than one antibiotics at 100 mg/ml concentration.

Table II
Results of the experiment of curing

Before Curing					
Code No.	Ampicillin	Streptomycin	Tetracycline	Chloramphenicol	Gentamicin
ST 67	Am (500)	Sm (500)	Tc (100)	Cm (500)	Gm (< 10)
After Curing					
ST 67	Am (50)	Sm (300)	< Tc (100)	Cm (< 300)	GM (< 10)

Key:

Am = Ampicillin
Sm = Streptomycin
Tc = Tetracycline
Cm = Chloramphenicol
Gm = Gentamicin

Table III
Resistance pattern (mg/ml) of Donor, Recipient and Transconjugant
in experiment of conjugation with *Salmonella typhi*

Code No.	Ampicillin	Streptomycin	Tetracycline	Chloramphenicol	Gentamicin
ST 67	Am (500)	Sm (500)	Tc (100)	Cm (500)	Gm (< 10)
<i>E coli</i>	Am (50)	Sm (50)	Tc (50)	Cm (25)	Gm (< 10)
Transconjugant	Am (500)	Sm (300)	Tc (100)	Cm (300)	Gm (<10)

Key:

Am = Ampicillin
Sm = Streptomycin
Tc = Tetracycline
Cm = Chloramphenicol
Gm = Gentamicin

Table IV
Resistance pattern of Donor, Recipient and Transconjugant
in the experiment of conjugation with *S. para typhi A*

Code No.	Ampicilli	Streptomycin	Tetracycline	Chloramphenicol	Gentamicin
STA 1	Am (500)	Sm (500)	Tc (500)	Cm (500)	Gm (< 10)
<i>E coli</i>	Am (500)	Sm (100)	Tc (50)	Cm (100)	Gm (< 250)
Transconjugant	Am (500)	Sm (300)	Tc (300)	Cm (300)	Gm (< 250)

Key:

Am = Ampicillin
Sm = Streptomycin
Tc = Tetracycline
Cm = Chloramphenicol
Gm = Gentamicin

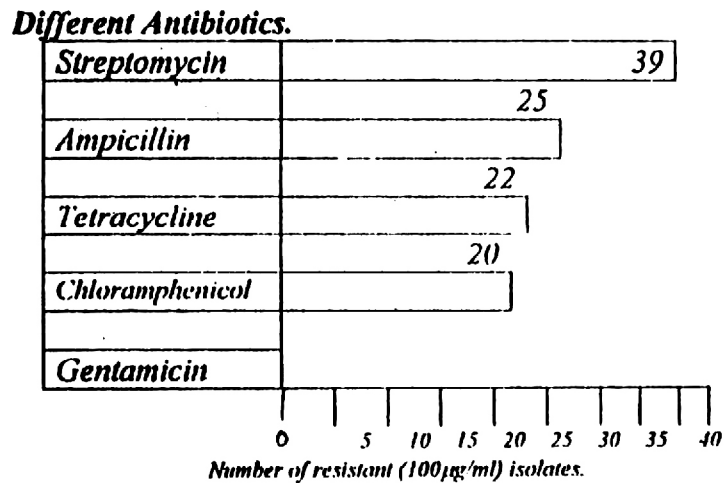


Figure 1: Resistance shown by *Salmonella* spp. to different antibiotics.

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